

VAE, EVAVC & EVCL Emulsion Product Catalogue

VAE, EVAVC & EVCL 乳液产品目录



Adhesives
接 着



Paint & Coating
涂 料



Construction &
Building
建 筑

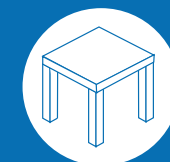


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Company Background 公司简介

Dairen Chemical Corporation (DCC) was established in year 1979 and it has been operated based on the business philosophy of "Integrity as the business foundation", "Customer-oriented product and service" and "Being creative and innovative". DCC is one of the members of the Chang Chun group.

DCC started producing vinyl acetate monomer in year 1983. Then, it began conducting research on the possible downstream products of vinyl acetate monomer. After five years, it had successfully commercialized the production of vinyl acetate-ethylene (VAE) emulsion via its own technology. Furthermore, in year 1996, DCC adopted spray drying technology to convert the VAE emulsion into redispersible VAE powder, which is a widely used additive in many construction applications. Over the decades, the continual research & development and quality improvement effort has made the DCC product to gain international recognition.

In year 2004, apart from VAE, DCC starts producing vinyl acetate-vinyl chloride-ethylene (EVAVC) emulsion and vinyl chloride-ethylene (EVCL) emulsion.

DCC has four manufacturing facilities worldwide which are located at Kaohsiung (Taiwan), Johor (Malaysia), Jiangsu (China) and Singapore. Besides, DCC has established three technical centers that are located at Kaohsiung (Taiwan), Johor (Malaysia), Jiangsu (China) to provide technical support to our customer. Our products have been sold globally, for instance, to China, Japan, Korea, India, Southeast Asia, Australia, America, Europe, etc. We always strive for continual improvement and innovation to provide better product and immediate service to our customer.

大连化学工业股份有限公司 (DCC) 成立于 1979 年，一直以来，秉持著诚信立业、顾客至上、创意革新为企业的经营理念。DCC 是长春集团的其中一个成员。

大连化学工业股份有限公司 1983 年开始生产醋酸乙烯酯，即进行开发下游衍生物，于 1985 年开始生产自行开发的 VAE 乳液。由于经过多年持续不断地努力，并致力于研究开发与品质改善，使得产品在市场上大获好评。于 1996 年开始将 VAE 乳液进行喷雾干燥生产 VAE Powder。

2004 年起开始生产 EVAVC 及 EVCL 乳液。DCC 於世界有四处生产基地，台湾高雄大社厂、马来西亚柔佛州厂、中国江苏仪征厂和新加坡厂。

此外，DCC 设有三个技术服务中心，分别位于台湾高雄、马来西亚柔佛和中国江苏仪征，提供技术服务给予我们的顾客。销售市场遍及世界各地，包括中国、日本、韩国、印度、东南亚、澳洲、美洲、欧洲等等。为配合全球市场上的需求，公司走向国际化，继续提供客户更具竞争力的产品品质、更迅速即时的服务。



Taiwan 台湾高雄厂



Singapore 新加坡厂



Malaysia 马来西亚厂



China 中国江苏厂



Background of Polymer Emulsion

Vinyl acetate-ethylene emulsion (VAE) is a milky white, non-toxic, low-odor, non-flammable/ explosive emulsion that can be transported easily. Its main functions are to increase the adhesion, flexibility, water resistance of the final product. The VAE emulsion can be applied on a wider range of substrate as compared to polyvinyl acetate. One of its noticeable applications is as the adhesive between PVC sheet and other substrates.

Besides, the VAE emulsion can be used as the adhesive of various types of plastic materials and porous substrates. Examples of plastic materials are polyvinylidene chloride, cellulose acetate, polystyrene whereas examples of porous substrates are paper, wood, woven, fiber board and polyurethane foam. Furthermore, the VAE emulsion is widely used in construction industry as the adhesion promoter and water-resistance promoter of cement admixture.

To enhance the quality of the environment, maintaining a safe and healthy living environment, to ensure sustainable development of enterprises, the company adopts environmental protection, safety and health policies, and has great commitment in improving the production and applications of the existing products, designing and developing new process and product, enhancing the value and competitiveness of the product, providing satisfactory product to customer, in order to achieve the company philosophy.

VAE emulsion is a copolymer of vinyl acetate and ethylene and it is a strong water-borne adhesive with the features of

- (1) High initial adhesion strength
- (2) Excellent heat resistance
- (3) Good water resistance
- (4) Good alkaline resistance

- (5) Good thickening response
- (6) Safe for operation

On the other hand, the vinyl acetate-vinyl chloride-ethylene (EVAVC) emulsion and vinyl chloride-ethylene (EVCL) emulsion have the features of

- (1) Fire-retardant
- (2) Excellent water resistance
- (3) Excellent solvent resistance
- (4) Self-cross-linkable

Main applications are:

(1) Adhesives:

- **Furniture decoration:** Excellent heat resistance, high adhesion strength, fast setting
- **Parquet:** Fast setting, excellent heat resistance
- **Automotive industry:** Excellent heat resistance, high adhesion strength
- **Paper and packaging board:** Fast setting, excellent processing property
- **Non-woven:** Excellent anti-aging property, low formaldehyde emission
- **Woodworking:** Excellent bonding strength

(2) Paint and coating:

- Low odor, low volatile organic compound (VOC) emission, superior hiding power, excellent scrub resistance, good filler and additive compatibility

(3) Construction & Building:

- Flexible, improved water resistance

乳液简介

VAE 为不含增塑剂乳白色水溶液醋酸乙烯 - 乙烯共聚物乳液，无毒、低气味、不易燃 / 易爆，携带运输方便。主要用于胶粘剂、涂料、建筑和纸质加工行业，具有有多优良的性质；比聚醋酸乙烯乳液应用于接着材质更广泛，尤其适用在 PVC 皮之接着；此外亦可作为各类塑胶模与多孔性基材之接着剂，塑胶膜如聚偏氯乙烯、醋酸纤维素、聚苯乙烯，多孔性基材如纸、木材、绵布、纤维板及 PU 泡绵。亦可广泛用于土木建材方面作为水泥接着剂之增强剂、水泥防水剂。

为全面提升环境品质，维护安全健康的生活环境，确保企业永续发展，本公司诉求环保、安全、卫生的政策，并持续致力于现有产品应用与制程改善、新制程与产品开发等，提高产品的附加价值与竞争力，提供顾客满意的产品，以实现公司经营理念。

VAE 乳液为醋酸乙烯及乙烯之共聚物，是一种强力水性接着剂，具有多项特长

- (1) 初期接着性高
- (2) 优良耐热潜变性
- (3) 耐水性佳
- (4) 耐碱性佳
- (5) 增黏性高
- (6) 安全的作业性

醋酸乙烯 - 乙烯 - 氯乙烯共聚物 (EVAVC) 乳液和乙烯 - 氯乙烯共聚物 (EVCL) 乳液的特色为

- (1) 阻燃
- (2) 耐水性佳

- (3) 耐溶剂性佳
- (4) 自交联性佳

主要用途：

(1) 接着

家具装潢：良好的热安定性、优良的接着强度、快速的粘结速度。

拼花地板：快速粘合速度、耐热性佳。

汽车产业：耐热性佳、良好的接着力。

包装及纸张接着：凝结速度快、良好维持能力、良好加工性。

无纺布与不织布：优良的耐老化特性、低甲醛产品。

木工胶：优良的接着强度。

(2) 涂料

低气味、低有机挥发份、杰出遮蔽力、优良的耐刷洗性能、易于调整添加剂之组成。

(3) 建筑

良好的弹性、良好的耐水性。





Wood adhesive
木材接着



PVC adhesive
PVC 接着



Paper adhesive
纸类接着



Packaging adhesive
包装接着



Aluminium adhesive
铝箔接着



Car decorative veneer adhesive
汽车饰板接着

Grade 规格	Stabilization system 稳定体系	Typical Properties 乳液特性						Special feature 产品特性	Application 应用领域				Grade 规格
		Solid content 固形份 (wt%)	Viscosity 黏度 (cP)	Spindle/ RPM 转子 / 转速	pH 酸碱值	Tg 玻璃转化温度 (℃)	MFFT 最低成膜温度 (℃)		General adhesive 一般接着	Cigarette adhesive 卷烟胶	Wood adhesive 木材接着	Nonwoven/ Carpet 无纺纸接着 / 地毯背胶	
Vinyl acetate-ethylene (VAE) emulsion 醋酸乙烯 – 乙烯共聚物													
DA-100	P	55.0	1100-1600	3/60	4.5-6.5	0	0	Superior bonding for a wide range of application. Provides soft and flexible film for various porous and non-porous substrates, without further additive required, and especially useful for paper and plastic packaging. 对各种多孔和非多孔基材有良好的接着，在无需添加任何助剂亦能给予柔软和柔性膜。拥有良好的耐热和耐水性质，特别适合用于纸张和塑料包装接着。	○				DA-100
DA-100L	P	54.5	500-1000	3/60	4.5-6.5	0	0		○				DA-100L
DA-101	P	55.0	1500-2500	4/60	4.5-6.5	0	0		○		○	○	DA-101
DA-102	P	55.0	2500-3700	4/60	4.5-6.5	0	0		◎		○	○	DA-102
DA-102H	P	55.0	3500-4500	4/60	4.5-6.5	0	0	High viscosity version of DA-101. Suitable for difficult-to-adhere substrate surface adhesion. 性质同 DA-101，黏度较高之乳液。拥有良好的耐热与耐水性质，适用於难贴表面基材的接着。	◎		○	○	DA-102H
DA-103	P	60.0	1500-2500	4/60	4.5-6.5	0	0	High solid content version of DA-101. Good heat and water-resistance for adhesion. 性质同 DA-101, 固形份较高之乳液。拥有良好的耐热与耐水性质，适用於接着。	○		○		DA-103
DA-103H	P	60.0	3200-4200	4/60	4.5-6.5	0	0	High solid content and viscosity version of DA-103. Good heat and water-resistance for adhesion. 性质同 DA-103，黏度较高之乳液。拥有良好的耐热与耐水性质，适用于接着。	○		○		DA-103H
DA-103I	P	58.0	850-1050	2/6	4.0-7.0	0	0	Dispersion combines high solid content and low viscosity and allows the addition of high filler loading. Suited in carpet backing application where a medium/soft hand is required. 黏度较低之乳液。拥有良好的粉料相容性，适用于地毯背胶，手感中等。				◎	DA-103I
DA-104	P	55.0	2000-3000	4/60	4.5-6.5	-10	0	Low Tg. Flexible adhesive specifically tailored for low temperature packaging application. 低玻璃转化温度。柔软性质的胶黏剂专为低温包装接着。	◎				DA-104
DA-105	P+S	50.0	4000-6000	4/20	4.5-6.5	0	0	Low free monomer. Suitable for adhesion of difficult substrate such as lacquered surface, foil or plastic film. Suitable for cigarette adhesive application. 低残单。适于难贴表面基材（如漆表面、金属箔或塑料薄膜）的接着。也适用于卷烟胶。	◎				DA-105
DA-107	P+S	65.0	1000-2500	4/60	4.5-6.5	0	0	High solid content version of DA-101. Suitable for adhesion and construction use. 性质同 DA-101，固形份较高之乳液。可用于接着与建筑应用。	○				DA-107
DA-107L	P+S	65.0	500-1500	3/60	4.5-6.5	0	0	High solid content and low viscosity version of DA-107. Suitable for adhesion and construction use. 性质同 DA-107，黏度较低之乳液。拥有良好的耐热与耐水性质，可用于接着与建筑应用。	○				DA-107L

P = PVOH 聚乙烯醇, C = Cellulose derivative 纤维素, S = Surfactant 乳化剂

◎ = Highly recommended 强烈推荐, ○ = Recommended 推荐



Wood adhesive
木材接着



PVC adhesive
PVC 接着



Paper adhesive
纸类接着



Packaging adhesive
包装接着



Aluminium adhesive
铝箔接着



Car decorative veneer adhesive
汽车饰板接着

Grade 规格	Stabilization system 稳定体系	Typical Properties 乳液特性						Special feature 产品特性	Application 应用领域				Grade 规格	
		Solid content 固形份 (wt%)	Viscosity 黏度 (cP)	Spindle/ RPM 转子 / 转速	pH 酸碱值	Tg 玻璃转化温度 (°C)	MFFT 最低成膜温度 (°C)		General adhesive 一般接着	Cigarette adhesive 卷烟胶	Wood adhesive 木材接着	Nonwoven/ Carpet 无纺纸接着 / 地毯背胶		
Vinyl acetate-ethylene (VAE) emulsion 醋酸乙烯 – 乙烯共聚物														
DA-108	P+S	72.0	1500-2500	4/60	4.5-6.5	0	0	High solid content version of DA-101. Good heat and water-resistance for adhesion. 性质同 DA-101，固形份较高之乳液。	○					DA-108
DA-108H	P+S	72.0	2500-5000	4/60	4.5-6.5	0	0	High solid content and viscosity. Good heat and water-resistance for adhesion. 性质同 DA-108，黏度较高之乳液。拥有良好的耐热与耐水性质，适用于接着。	○					DA-108H
DA-111	P	55.0	1500-2500	4/60	4.5-6.5	14	-1	High Tg. Suitable for paper and packaging adhesion and construction use. 高玻璃转化温度。可用于接着与建筑应用。	○					DA-111
DA-125	P+S	55.0	2500-3500	4/60	4.5-6.5	-8	0	Specifically tailored for low temperature packaging application. 适用于低温接着。	◎					DA-125
DA-128	P+S	60.0	5000-7500	5/20	4.5-6.5	-10	0	Low Tg. Flexible adhesive specifically tailored for low temperature packaging application. 低玻璃转化温度。柔软性质的胶黏剂专为低温包装接着。	◎					DA-128
DA-128H	P+S	60.0	7500-13000	5/20	4.5-6.5	-10	0	Low Tg. Flexible adhesive specifically tailored for low temperature packaging application. Suitable for difficult substrate surface adhesion. 低玻璃转化温度。柔软性质的胶黏剂专为低温包装接着。拥有良好的耐热与耐水性质，适用于难贴表面基材的接着。	◎					DA-128H
DA-140L	P	54.5	500-1000	3/60	4.5-6.5	0	0	Low free monomer, VOC and formaldehyde. Adhesive with superior bonding that is specifically formulated as cigarette adhesive. 极低残余单体及 VOC，无甲醛，由良好的快干性，适用于高速卷烟纸接嘴、搭口以及包装胶。		◎				DA-140L
DA-141	P	55.0	1500-2500	4/60	4.5-6.5	0	0			◎				DA-141
DA-142	P	55.0	2500-3700	4/60	4.5-6.5	0	0			◎				DA-142
DA-143	P	60.0	1500-2500	4/60	4.5-6.5	0	0			◎				DA-143
DA-144	P	55.0	2000-3000	4/60	4.5-6.5	-15	0			◎				DA-144
DA-146	P	55.0	1800-2700	3/20	4.5-6.5	15	2			◎				DA-146
DA-147	P+S	55.0	1800-2700	3/20	4.5-6.5	15	1			◎				DA-147
DA-161	P	55.5	1500-3000	4/60	4.8-6.0	5	0	Suitable for roller or high speed nozzle application. Good redispersibility for easy cleaning and good heat resistance. 适用于易清洗胶。	◎					DA-161

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◎ = Highly recommended 强烈推荐，○ = Recommended 推荐



Wood adhesive
木材接着



PVC adhesive
PVC 接着



Paper adhesive
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Packaging adhesive
包装接着



Aluminium adhesive
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Car decorative veneer adhesive
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Grade 规格	Stabilization system 稳定体系	Typical Properties 乳液特性						Special feature 产品特性	Application 应用领域				Grade 规格
		Solid content 固形份 (wt%)	Viscosity 黏度 (cP)	Spindle/ RPM 转子 / 转速	pH 酸碱值	Tg 玻璃转化温度 (°C)	MFFT 最低成膜温度 (°C)		General adhesive 一般接着	Cigarette adhesive 卷烟胶	Wood adhesive 木材接着	Nonwoven/ Carpet 无纺纸接着 / 地毯背胶	
DA-201	P	55.0	1300-2200	4/60	4.0-6.5	0	0	Provides soft and flexible film for various porous and non-porous substrates, without further additive required. Especially suited to be used in the textile industries that the product needed in white-based color. 对各种多孔和非多孔基材有良好的接着，在无需添加任何助剂亦能给予柔软和柔性膜。拥有良好的耐热与耐水及高温低黄变性质，适用于接着。	○				DA-201
DA-201E	P	55.0	2000-2800	4/20	4.0-6.0	0	0		○				DA-201E
DA-202	P	55.0	2000-3500	4/60	4.0-6.5	0	0		○				DA-202
DA-202E	P	55.0	3000-4000	4/20	4.0-6.0	0	0		○				DA-202E
DA-265	C+S	65.0	1000-2000	4/60	4.5-6.5	-25	0	Low Tg. Suitable for adhesion of difficult substrate such as lacquered surface, foil or plastic film and pressure sensitive adhesive application. 低玻璃转化温度。适用于 PVC、纸、OPP 等基材接着。也可用于塑料层与压敏接着剂。	◎				DA-265
DA-265H	C+S	65.0	2000-3000	4/60	4.5-6.5	-25	0	High viscosity version of DA-265. Suitable for adhesion of difficult substrate such as lacquered surface, foil or plastic film and pressure sensitive adhesive application. 性质同 DA-265，黏度较高之乳液。低玻璃转化温度。适用于 PVC、纸、OPP 等基材接着。也可用于塑料层与压敏接着剂。	◎				DA-265H
DA-266	C+S	65.0	1000-2000	4/60	4.5-6.5	-30	0	Low Tg. Suitable for adhesion of difficult substrate such as lacquered surface, foil or plasticfilm and pressure sensitive adhesive application. 低玻璃转化温度。适用于 PVC、纸、OPP 等基材接着。也可用于塑料层与压敏接着剂。	◎				DA-266
DA-310	P+S	60.0	1000-2000	4/60	3.5-5.5	0	0	Suitable for metalized substrates adhesion application. 适用于金属材料包装接着。	◎				DA-310
DA-342	S	52.0	50-500	2/60	4.0-7.0	0	0	Low formaldehyde content. Suitable used as nonwoven binders. Medium/soft hand feel. 低甲醛含量。适用于无纺纸接着与纤维处理。手感中等粗硬。				◎	DA-342
DA-346	S	52.0	50-500	2/60	4.0-7.0	-15	0	Low formaldehyde content. Suitable used as nonwoven binders. Soft hand feel. 低甲醛含量。适用于无纺纸接着与纤维处理。手感柔软。				◎	DA-346
DA-500	P+S	54.5	500-1000	3/60	4.5-6.5	0	0	Suitable for high speed packaging, good heat and water resistance. 拥有良好的耐热与耐水性性质，适用于高速包装接着。	○			◎	DA-500
DA-502	P+S	54.5	3000-4000	4/60	4.5-6.5	0	0	Good heat and water resistance, especially suitable for general adhesive application. 拥有良好的耐热与耐水性性质，适用于一般接着。	○				DA-502
DA-511	P	55.0	1500-2500	4/60	4.5-6.5	15	2	High Tg, suitable for high speed packaging, excellent mechanical stability and fast setting speed. 高玻璃转化温度，拥有良好的耐热、耐水、机械稳定性与快干燥速率性质。适用于高速包装接着。	◎		○		DA-511
DA-524	P	55.0	1500-2500	4/60	4.5-6.5	20	3	High Tg, suitable for general adhesive, good heat resistance and good water sensitivity. 高玻璃转化温度，拥有良好的耐热与水溶性性质。适用于一般接着。	○		◎		DA-524

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PVC adhesive
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Paper adhesive
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Car decorative veneer adhesive
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Grade 规格	Stabilization system 稳定体系	Typical Properties 乳液特性						Special feature 产品特性	Application 应用领域				Grade 规格
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Vinyl acetate-ethylene (VAE) emulsion 醋酸乙烯 – 乙烯共聚物

DA-652	P	55.5	1500-20000	4/30	4.0-7.0	0	0	Good heat and water resistance. 拥有良好的耐热与耐水性质。	◎				DA-652
DA-691	P+S	54.5	100-500	2/60	5.0-7.5	0	0	Good heat and water resistance, especially suitable for D4 grade woodworking adhesive. 拥有良好的耐热与水溶性性质。适用于集成材用双组份木工胶。可达欧洲 标准 EN204 D4 级。	○		◎		DA-691

Ethylene-vinyl acetate-vinyl chloride (EVAVC) emulsion 醋酸乙烯 – 乙烯 – 氯乙烯共聚物

DA-901	P	50.0	1000-3000	3/60	5.0-7.0	0	0	General adhesive with fire-retardant property, good bonding strength as D4 grade woodworking adhesive. 具防火性质的一般接着剂。适用于集成材用双组份木工胶。可达欧洲标 准 EN204 D4 级。	○		◎		DA-901
DA-903	C+S	50.0	10-200	2/60	5.0-7.0	20	30	Self-crosslinkable emulsion with fire-retardant property. Suitable for textile, non-woven and carpet backing applications. 具防火性质的自交联接着剂。适用于纺织、无纺布的接着与地毯背胶。				◎	DA-903
DA-905	C+S	50.0	10-100	2/60	5.0-7.0	30	40	General adhesive with fire-retardant property, extensive use in textile applications, excellent bonding strength as D4 grade woodworking adhesive. 具防火性质的一般接着剂。适用于纺织的接着与纤维处理。适用于集成 材用双组份木工胶。可达欧洲标准 EN204 D4 级。			◎	◎	DA-905
DA-908	P	50.0	200-1200	3/60	4.5-6.5	25	30	General adhesive with fire-retardant property, excellent bonding strength as D4 grade woodworking adhesive. 具防火性质的一般接着剂。适用于集成材用双组份木工胶。可达欧洲标 准 EN204 D4 级。	○		◎	○	DA-908

Ethylene-vinyl chloride (EVCL) emulsion 乙烯 – 氯乙烯共聚物

DA-910	S	50.0	1-50	1/60	6.0-8.0	0	0	Low Tg, self-crosslinkable emulsion with fire-retardant property. Suitable for non-woven and carpet backing applications. 低玻璃转化温度。具防火性质的自交联接着剂。适用于无纺布的接着、 纤维处理与地毯背胶。可应用于汽车内衬接着。				◎	DA-910
DA-930	S	50.0	1-50	1/60	7.0-8.0	30	40	High Tg, self-crosslinkable emulsion with fire-retardant property. Suitable for non-woven and carpet backing applications. 高玻璃转化温度。具防火性质的自交联接着剂。适用于无纺布的接着、 纤维处理与地毯背胶。可应用于汽车内衬接着。				◎	DA-930

P = PVOH 聚乙烯醇, C = Cellulose derivative 纤维素, S = Surfactant 乳化剂

◎ = Highly recommended 强烈推荐, ○ = Recommended 推荐

Water-based paint —Towards a greener and more sustainable society

The increased society awareness on safer and healthier painting working environment has shifted the use of solvent-based paint to water-based paint because the former emits relatively high concentration of volatile organic compounds (VOC) during the paint drying process.

For water-based interior paint, it is essential to achieve even lower VOC emission because good indoor air ventilation is not always guaranteed. The presence of high VOC concentration is often related to the detection of strong unpleasant smell, which is a nuisance to the users. Besides, long-term exposure to high VOC concentration may pose a threat to the health of the painters and residents.

DCC VAE emulsion—Your solution to good paint performance and low VOC emission

The most common sources of VOC in conventional water-based interior paint are coalescing agent and solvent. Removing these additives from paint formulation can reduce the VOC emission, but this will adversely affect the paint performance, for instance, leading to poor scrub resistance.

To resolve this conflict, Dairen Chemical Corporation (DCC) has developed a series of vinyl acetate-ethylene(VAE) emulsion that can help the paint formulator to attain a trade-off between good paint performance and low VOC emission.

Features of DCC VAE emulsion

Our APEO-free VAE emulsion has a minimum film forming temperature of 0°C. This permits good film formation at ambient temperature without the addition of coalescing agent.

Furthermore, the glass transition temperature of our VAE emulsion is in a range of 6-13°C, thus providing good scrub resistance to the paint.

Using our VAE emulsion to formulate paint can potentially reduce the dosage of titanium dioxide due to the inherent good hiding power of the VAE film.

It is proven that the commercial paint formulated from our VAE emulsion gives very low VOC emission.

水性涂料—迈向更环保，更可持续发展的社会

基於安全和健康的考量，现今社会倾向于使用水性涂料，因为溶剂型涂料在干燥过程中会释放出相对较高浓度的挥发性有机化合物（VOC）。

用於室内的水性涂料应该释放更加低的 VOC 因为不是每一个室内都拥有良好的通风系统。高 VOC 浓度往往会导致难闻气味的形成，进而造成漆匠的工作困扰。此外，长期暴露於高 VOC 浓度中可能对漆匠和居民的健康构成威胁。

大连化工 VAE 乳液 – 让您兼具良好的涂料性能和低 VOC 释放

基本上，成膜助剂和溶剂是常规水性内墙涂料里的 VOC 来源。虽然将这些添加剂从涂料配方中除去可以降低 VOC 的释放量，但是却会对涂料的性能产生不利的影响，例如，导致涂料的耐刷洗性能变差。

要解决这个矛盾，大连化学工业（DCC）开发了一系列醋酸乙烯-乙烯共聚物乳液（VAE），可以帮助涂料配方设计师在优越的涂料性能和低 VOC 释放之间取得一个良好的平衡。

大连化工 VAE 乳液的特色

大连的 VAE 乳液不含 APEO。此外，乳液的最低成膜温度为 0°C，在不加入成膜助剂的情况下，也能达到良好的成膜效果。

大连的 VAE 乳液的玻璃化转变温度是 6-13°C，使得涂料具备良好的耐刷洗性能。通过使用大连的 VAE 乳液，可以减少涂料配方中二氧化钛的添加量，因为大连 VAE 干膜拥有良好的遮蔽力。分析显示采用大连 VAE 乳液制成的水性涂料所释放的 VOC 量相当的低。

Benefits of using DCC VAE emulsion to formulate paint

- 1) APEO-free emulsion.
- 2) Good scrub resistance.
- 3) Low VOC emission.
- 4) Reduce titanium dioxide dosage.
- 5) Good film formation without the addition of coalescing agent.



使用 VAE 乳液水性内墙涂料的优点

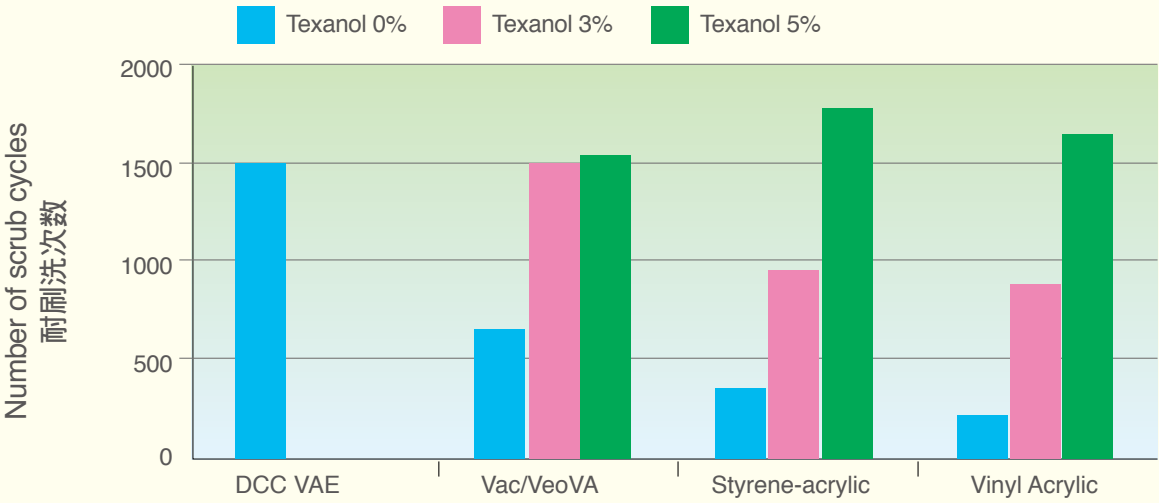
- 1) 乳液不含 APEO
- 2) 耐刷洗性能佳
- 3) 低 VOC 释放
- 4) 降低二氧化钛的添加量
- 5) 无需添加成膜助剂也能达到良好的成膜效果



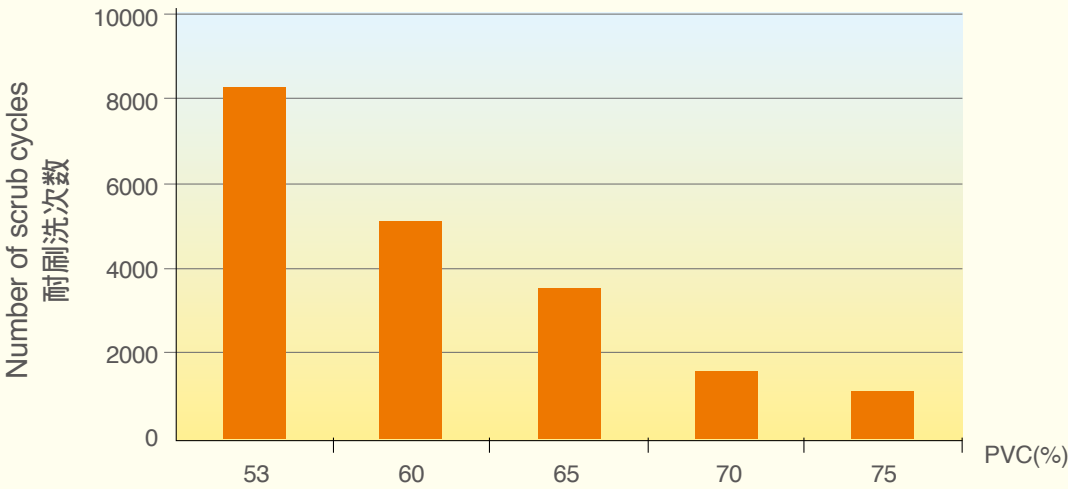
Choices of DCC VAE emulsion for water-based interior paint
适用于水性内墙涂料的大连化学工业 VAE 乳液

Grade 规格	Stabilization system 稳定体系	Typical properties 乳液特性						Recommended pigment volume concentration of water-based interior paint 建议用于水性内墙 涂料之颜料体积浓度		
		Solid content 固形份 (wt%)	Viscosity 黏度 (cP)	Spindle/ RPM 转子 / 转速	pH 酸碱值	Tg 玻璃 转化温度 (°C)	MFFT 最低 成膜温度 (°C)	High 高	Medium 中	Low 低
DA-371	P+S	55.0	500-1500	3/60	4.5-6.5	10	0	◎		
DA-381	P+S	55.0	100-1000	2/20	4.5-6.5	10	0		◎	
DA-382	P+S	55.0	100-700	2/20	4.0-6.5	8	0	◎		◎
DA-391	S	55.0	100-700	2/20	4.0-6.0	8	0	◎		◎

P = Protective colloid 保护胶体 , S = Surfactant 乳化剂



DCC VAE emulsion provides paint with good scrub resistance without the need of adding coalescing agent.
大连 VAE 乳液在无需添加成膜助剂的情况下也能赋予涂料优越的耐擦洗性能。



Paint formulated from DCC VAE emulsion shows excellent scrub resistance even at high PVC.
大连 VAE 乳液所制成的涂料在高 PVC 配方中依然拥有良好的耐擦洗性能。

Dairen Chemical Corporation (DCC) produces a variety of vinyl acetate-ethylene (VAE) emulsion that can be used in different construction applications such as primer, cement admixture, two-component cementitious waterproofing membrane, two-component cementitious tile adhesive and joint compound. Following are the benefits of using DCC VAE emulsion to formulate

Primer

- Reduce water permeation to base substrate
- Enhance the adhesion strength of mortar applied on base substrate

Cement admixture

- Increase the adhesion strength of mortar

Two-component cementitious waterproofing membrane (CWM)

- Improve the flexibility of CWM
- Enhance the waterproofing effect of CWM
- Can be applied on various type of substrate

Two-component cementitious tile adhesive

- Improve the adhesion strength of tile adhesive

Joint compound

- High filler loading capacity
- Improve the flexibility of joint compound

大连化学工业（DCC）生产的醋酸乙烯 - 乙烯共聚物乳液（VAE）适用于不同的建筑应用领域，例如吸水调整材、砂浆添加剂、双组份防水浆料、双组份瓷砖粘结剂和填缝材。使用 DCC VAE 乳液的优点为

吸水调整材

- 减少水渗透基材
- 增加砂浆粘结于基材的强度

砂浆添加剂

- 加强砂浆的粘结强度

双组份防水浆料

- 提升防水浆料的柔软度
- 防水性佳
- 可以涂布于不同种类的基材表面

双组份瓷砖粘结剂

- 提高粘结强度

填缝材

- 填料相容性佳
- 提升填缝材的柔软度

Choices of DCC VAE emulsion for construction application
适用于建筑业的大连化学工业 VAE 乳液

Grade 规格	Stabilization system 稳定体系	Typical properties 乳液特性						Application 应用领域				
		Solid content 固形份 (wt%)	Viscosity 黏度 (cP)	Spindle/ RPM 转子 / 转速	pH 酸碱值	Tg 玻璃转 化温度 (℃)	MFFT 最低 成膜温度 (℃)	Primer 吸水调 整材	Cement admixture 砂浆添 加剂	Water- proofing (2K) 防水浆料 (双组份)	Tile adhesive (2K) 瓷砖粘 结剂	Joint compound 填缝材
DA-117	C+S	65.0	500-1500	3/60	4.5-6.5	0	0	◎	◎	◎		◎
DA-125	P+S	55.0	2500-3500	4/60	4.5-6.5	-8	0			◎		
DA-180L	P+S	54.5	500-1000	3/60	4.5-6.5	0	0		◎	◎	◎	
DA-183	P+S	55.0	1500-2500	4/60	4.5-6.5	0	0			◎		
DA-187	P+S	55.0	200-2000	3/60	4.0-6.0	3	0	◎	◎	◎		

P = PVOH, C = Cellulose derivative, S = Surfactant

P = 聚乙烯醇, C = 纤维素, S = 乳化剂



In general, VAE emulsion is considered safe for its intended use. However, as this class of products always contains a minor amount of residual vinyl acetate monomers (less than 0.1%), adequate ventilation is recommended in rooms whenever VAE emulsion is handled. Generally when working with VAE emulsions, protective clothes, gloves and also goggles should be used. Splashes of VAE emulsion on the skin or into the eyes should immediately be removed by irrigation with clean water.

VAE emulsion should be stored in sheltered areas at temperatures between 5 and 40°C; do never store under freezing conditions. Soft lumps or skins may tend to be formed during transportation and storage. Thus, filtration is recommended before use. Adequate precautionary measures are needed if the VAE emulsion is not fully transferred at once to prevent the possible formation of lumps and skins. Opened containers should be closed soon to prevent skinning.

Ensure the cleanliness of equipment and facilities that are used to transfer the VAE emulsion that kept in original containers to other tanks. Additional preservative is required to prevent microbial attack. Shelf life of VAE emulsion is minimum 6 months, provided the product is stored in original, unopened containers under proper conditions. Stirring is required before utilizing the VAE emulsion if it has been stored for a long time. Finished products compounded from VAE polymer emulsions usually also require preservation.

Comprehensive instructions of safety notes are provided in the product corresponding technical data sheet (TDS) and safety data sheet (SDS) which are available on Dairen official website (www.dcc.com.tw) or upon request from Dairen sales officers.

一般而言，VAE 乳液使用上被认定很安全，但这些产品含少量的残存醋酸乙烯单体（少于 0.1%），建议使用 VAE 乳液时处于通风的空间，并且应该穿著防护措施，因 VAE 乳液是一种微酸之产品，操作人员应佩戴手套与护目镜，若 VAE 乳液喷溅至皮肤或眼睛，需以大量之清水冲洗。

VAE 乳液应该储存在 5°C～40°C 的阴凉环境中，避免低温冷冻存放。乳液在储存或运输过程中可能形成结皮或软结块。因此，我们推荐在使用之前先对其进行过滤。未使用完的产品应使用后应立即旋紧产品包装盖，避免产生结皮或软结块现象。

乳液转移或者分装过程中所使用的储槽和运输管道系统应保持充分的清洁。为避免乳液遭受微生物污染，则必须额外添加防腐剂。在适当的储存环境下，VAE 乳液在未开封的原装容器中的保质期为 6 个月。产品长时间储存后，使用前需搅拌均匀。使用 VAE 乳液制成的成品通常也需要进行必要的防腐措施。

详细的安全信息可向大连销售代表索取或者登陆大连网站 (www.dcc.com.tw) 下载产品技术资料表 (TDS) 和物质安全数据表 (SDS)。





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